

ATPL EASA STUDY GUIDE | 010 - AIR LAW

Chapter 20: Aerodrome Lighting & PAPI Systems

Syllabus Source: CAE Oxford (pp. 425-446) | Pass Target: >= 90% AviationExam

1. RUNWAY LIGHTING COLORS (MASTER PROGRESSION)

Lighting System	Color(s)	Location & Warning Progression
Runway Edge Lights	Variable WHITE	White, EXCEPT that on instrument runways the last 600 m (or 1/3, whichever is less) are YELLOW / AMBER to warn of runway end.
Runway Threshold Lights	GREEN	Unbroken transverse line of green lights showing in direction of approach. May include green wing bars.
Runway End Lights	RED	Unbroken transverse line of red lights showing in direction of take-off / rollout.
Runway Centerline Lights	WHITE, RED/WHITE, RED	White from threshold to 900 m from end; Alternating RED and WHITE from 900 m to 300 m from end; RED for the final 300 m.
Touchdown Zone (TDZ) Lights	WHITE barrettes	Extends from threshold for 900 m (or midpoint). Transverse rows of white barrettes.
Taxiway Edge / Centerline	BLUE edge / GREEN centerline	Taxiway edge lights are BLUE; Centerline lights are GREEN (flashing or alternate yellow/green in runway lead-on).

2. PRECISION APPROACH PATH INDICATOR (PAPI)

PAPI consists of a wing bar of 4 multi-lamp units on the left side of the runway. Standard glidepath: 3.0°:

- 4 White Lights: High (above 3° 30').
- 3 White, 1 Red: Slightly High (3° 10').
- 2 White, 2 Red: ON GLIDEPATH (3° 00' - 'Two red, two white, you're all right').
- 1 White, 3 Red: Slightly Low (2° 50').
- 4 Red Lights: Low (below 2° 30' - 'All red, you're dead').

3. VISUAL APPROACH SLOPE INDICATORS (VASIS & 3-BAR VASIS)

System Type	Visual Display Geometry & Pilot Guidance
Standard VASIS (2-Bar)	Upwind and downwind bars. Standard 3° glidepath. Red over White = On slope; Red over Red = Low; White over White = High.
3-Bar VASIS	Developed for wide-body aircraft with large eye-to-wheel height (e.g. B747). Uses middle bar as common bar. Flight crew of large aircraft fly upper glidepath (Bar 2 and 3).
T-VASIS	T-shaped light arrays on both sides of runway. Inverted 'T' indicates fly down (high); upright 'T' indicates fly up (low).

4. APPROACH LIGHTING SYSTEMS (ALS)

- Simple Approach Lighting System (SALS): Single row of lights extending at least 420 m from threshold with a 30 m crossbar at 300 m.

- Calvert Precision Approach System: Extends 900 m from threshold with 5 crossbars at 150 m intervals. Centerline lights and side row barrettes.
- Stop Bars: Unidirectional red lights embedded across taxiway at runway holding positions. Extinguished when ATC clearance is given.
- Runway Guard Lights: Pair of flashing yellow lights (wig-wags) located at taxiway/runway intersection to warn of approaching runway.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What color are runway centerline lights in the final 300 m of the runway?

- [A]: Alternate red and white
- [B]: White
- [C]: Red
- [D]: Yellow

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Centerline lights are white to 900 m, alternate red/white from 900 m to 300 m, and all RED for the final 300 m.

Q2: What is the PAPI indication for an aircraft on the correct glidepath?

- [A]: 3 white and 1 red
- [B]: 2 white and 2 red
- [C]: 4 white
- [D]: 1 white and 3 red

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Two red and two white indicates the aircraft is on the 3.0° glidepath.

Q3: What color are runway edge lights in the last 600 m of a precision instrument runway?

- [A]: Red
- [B]: Yellow / Amber
- [C]: Green
- [D]: Blue

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The last 600 m (or one third of runway length) of edge lights are yellow to warn of approaching runway end.

Q4: What color are taxiway edge lights and taxiway centerline lights?

- [A]: White edge and green centerline
- [B]: Blue edge and green centerline
- [C]: Yellow edge and blue centerline
- [D]: Green edge and white centerline

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under Annex 14, taxiway edge lights are blue and centerline lights are green.

Q5: What does a red stop bar light embedded across a taxiway indicate to a pilot?

- [A]: Proceed with caution
- [B]: Runway holding position: MUST NOT CROSS until light is extinguished by ATC
- [C]: Rapid exit taxiway
- [D]: De-icing pad entrance

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. An illuminated red stop bar is an absolute stop signal. Never cross an active red stop bar, even if verbal clearance was received.

Q6: On a precision instrument runway, what color are Touchdown Zone (TDZ) lights and how far do they extend?

- [A]: Green lights extending 300 m
- [B]: White barrettes extending from threshold for 900 m (or midpoint of runway)
- [C]: Alternating yellow and red lights for 600 m
- [D]: Blue lights for 1,200 m

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. TDZ lights consist of white barrettes extending from the threshold for 900 m (or to runway midpoint, whichever is less).